MOVING LESLIE MODELS ON CONSECUTIVE LOGARITHMIC WINDOWS: RIGID CRITICAL GEOMETRY, AUXILIARY REPAIR, EXACT OVERLAP SCALARIZATION, AND DATA-COMPATIBLE CAUSAL TESTS

Abstract. This version separates three questions that must not be conflated: static spectral completion, exact recurrence compatibility with the observed scalar sequence, and dynamical transport between overlapping windows. For a fixed Perron value b, K prescribed critical conjugate pairs, and a fixed auxiliary degree q, the repaired coefficient family has ambient degree $D = 2K + 1 + q$ and, at regular points, dimension $K + q$ and codimension $K + 1$, independently of q. For order-D recurrence windows displaced by s samples, the exact common-equation slice has generic local dimension $\max\{0, s - K - 1\}$. Hence the first generic one-dimensional repaired motion still occurs at $s_* = K + 2$. The repaired update is generally a nonlinear curve rather than the rigid affine line. If the first-new-value observation is transverse to this curve, that single scalar selects the local next coefficient vector and the remaining $K + 1$ new values are then fixed recursively.

The auxiliary multiplier coefficients are therefore not independent innovations under the fixed-(K, q, b) regular continuation hypothesis; they become hidden channels only if a completion, branch, q, or b is reselected using new information. Static Toeplitz completion alone does not establish a data-compatible moving realization. A recurrence-first, frequency-free membership test is consequently made primary. The numerical audit performed on 5 September 2026 is retained as a negative baseline for the then-current approximate pipeline, while exact conclusions are restricted to the hypotheses actually verified.

Contents

Remark 0.1. This document is a research notebook. The algebraic identities, the rigid overlap theorem, the repaired local-dimension theorem, and the local one-scalar continuation theorem are proved only under their stated rank, regularity, fixed-degree, fixed-Perron, and branch hypotheses. The proposed connection with consecutive windows of the Mertens function remains an experimental programme, not a proved arithmetic theorem. In particular, static spectral completion does not imply that the completed polynomial is the recurrence polynomial of the data, and no statement below proves a new estimate for the Mertens function or the Riemann hypothesis.

1. Purpose and change of viewpoint

The earlier static experiments asked whether one finite collection of fitted frequencies could be embedded in one nonnegative Leslie matrix. Allowing auxiliary eigenvalues made that static problem feasible in all forty reported configurations, but the result did not distinguish a coherent dynamical law from independent fitting.

The present formulation replaces independent static realizations by a sequence of rigid or fixed-degree repaired Leslie matrices attached to consecutive logarithmic windows. The object of study is therefore a non-autonomous product, or linear cocycle, rather than one fixed matrix. Linear cocycles and their long products are standard objects in dynamical-systems theory [**?**].

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Warning 1.1. The use of a time-dependent matrix product is not itself a new result. The potentially discriminating content is the simultaneous enforcement of rigidity, nonnegativity, continuity through the feasible set, and prediction of a future window without recalibration.

Proposition 1.2 (Three logically distinct layers)**.** *The following problems are distinct and must be tested separately.*

(1) Static spectral completion: *a prescribed critical core polynomial is multiplied by an auxiliary monic factor until a nonnegative Leslie characteristic polynomial is obtained.*
(2) Data-compatible realization: *the characteristic polynomial is the polynomial of an order-D recurrence actually satisfied by the scalar data window.*
(3) Dynamic transport: *two consecutive data-compatible realizations are joined using only the common past equations and a predeclared state rule.*

Static feasibility in the first sense does not imply either of the latter two properties.

Remark 1.3. For the arithmetic programme the logically clean direction is therefore

$$\text{data} \longrightarrow \text{recurrence polynomial} \longrightarrow \text{spectral factorization} \longrightarrow \text{Leslie positivity and transport.}$$

A frequency estimator may remain useful as an exploratory diagnostic, but it is not needed for the exact recurrence-first membership tests developed below.

2. Rigid moving Leslie matrices

Definition 2.1. For $m \geq 2$, a companion Leslie matrix is

$$L(a) = \begin{pmatrix} a_1 & a_2 & \cdots & a_m \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & & \ddots & 0 \end{pmatrix}, \qquad a_\ell \geq 0. \tag{2.1}$$

Its characteristic polynomial is

$$P(z;a) = z^m - \sum_{\ell=1}^{m} a_\ell z^{m-\ell}. \tag{2.2}$$

Fix $K \geq 1$ and set

$$m = 2K + 1. \tag{2.3}$$

For $b > 1$ and phases $\Omega = (\omega_1, \ldots, \omega_K)$, prescribe the roots

$$b, \qquad \sqrt{b}\, e^{\pm i\omega_1}, \ldots, \sqrt{b}\, e^{\pm i\omega_K}. \tag{2.4}$$

The unique monic polynomial with these roots is

$$P_{b,\Omega}(z) = (z-b) \prod_{k=1}^{K} \left(z^2 - 2\sqrt{b}\cos(\omega_k) z + b \right). \tag{2.5}$$

Writing

$$P_{b,\Omega}(z) = z^m - \sum_{\ell=1}^{m} a_\ell(b,\Omega) z^{m-\ell} \tag{2.6}$$

defines the coefficient map $A : (b, \Omega) \mapsto a(b, \Omega)$.

Proposition 2.2. *For fixed (b, Ω), the rigid Leslie coefficients in* (**??**) *are unique. A rigid nonnegative Leslie realization exists if and only if*

$$a_\ell(b,\Omega) \geq 0, \qquad 1 \leq \ell \leq m. \tag{2.7}$$

Proof. The roots in (**??**) determine the monic polynomial (**??**). Comparison with (**??**) determines every coefficient. The companion matrix is entrywise nonnegative exactly when its first-row coefficients satisfy (**??**). □

Definition 2.3. The rigid admissible phase set is

$$\mathcal{A}_{b,K} = \left\{\Omega \in \mathbb{T}^K : a_\ell(b,\Omega) \geq 0 \text{ for all } \ell\right\}. \tag{2.8}$$

Remark 2.4. The boundary of (**??**) is contained in the union of the coefficient zero sets

$$\partial\mathcal{A}_{b,K} \subseteq \bigcup_{\ell=1}^{m} \{\Omega : a_\ell(b,\Omega) = 0\}. \tag{2.9}$$

Equality need not hold without excluding redundant inequalities and singular intersections.

3. Windowed cocycle and accumulated phase

Let $W_0, W_1, \ldots$ be consecutive windows in logarithmic scale. Associate a rigid Leslie matrix

$$L_j = L(a_j), \qquad a_j = a(b_j, \Omega_j) \in \mathbb{R}_+^m. \tag{3.1}$$

The non-autonomous evolution is

$$v_{j+1} = L_j v_j, \tag{3.2}$$

with cocycle product

$$\Phi(J,0) = L_{J-1}\cdots L_1 L_0. \tag{3.3}$$

Remark 3.1. A moving sequence L_j has no mandatory spectral periodicity. This does not imply that every chosen trajectory is nonperiodic. Periodicity may reappear if the coefficients or phases are selected periodically.

If a locally matched branch has phase $\omega_{j,k}$ in window j, its formal accumulated phase is

$$\Theta_k(J) = \sum_{j=0}^{J-1} \omega_{j,k}. \tag{3.4}$$

The definition requires mode matching and phase unwrapping between adjacent windows. Without those operations, the unordered roots of successive polynomials do not determine a unique trajectory $k \mapsto \omega_{j,k}$.

4. Exact coefficient coordinates for the rigid critical spectrum

The general inverse spectral problem for Leslie matrices has an established characterization in the literature [**?**]. The results in this section concern the special rigid spectrum in (**??**) and its use as a moving-window coordinate system.

Put $r = \sqrt{b}$ and define

$$S_\Omega(w) = \prod_{k=1}^{K} \left(w^2 - 2\cos(\omega_k)w + 1\right) = \sum_{j=0}^{2K} d_j w^{2K-j}. \tag{4.1}$$

The polynomial is reciprocal, and hence

$$d_{2K-j} = d_j, \qquad d_0 = d_{2K} = 1. \tag{4.2}$$

Moreover,

$$P_{b,\Omega}(z) = (z - r^2) r^{2K} S_\Omega(z/r). \tag{4.3}$$

Proposition 4.1. *The coefficients in* (**??**) *satisfy*

$$a_\ell = r^\ell \left(r d_{\ell-1} - d_\ell\right), \qquad 1 \le \ell \le 2K, \tag{4.4}$$

$$a_m = r^{m+1} = b^{K+1}. \tag{4.5}$$

For $1 \le j \le K$*, the paired coefficients are*

$$a_j = r^j \left(r d_{j-1} - d_j\right), \tag{4.6}$$

$$a_{m-j} = r^{m-j} \left(r d_j - d_{j-1}\right). \tag{4.7}$$

Proof. Expand (**??**) and compare the coefficient of $z^{m-\ell}$ with (**??**). Equation (**??**) follows from (**??**). □

Corollary 4.2. *A rigid nonnegative Leslie realization exists if and only if*

$$d_j > 0 \quad \textit{and} \quad \frac{1}{r} \le \frac{d_j}{d_{j-1}} \le r, \qquad 1 \le j \le K. \tag{4.8}$$

Strict inequalities are equivalent to strict positivity of all paired coefficients.

Proof. Equations (**??**) and (**??**) give the two inequalities. If one of d_{j-1} and d_j vanished while both paired inequalities held, the other would also vanish. Propagation back to $d_0 = 1$ yields a contradiction. Thus feasibility forces positivity of all d_j for $0 \le j \le K$. □

Definition 4.3. On the region where $d_1, \dots, d_K > 0$, define

$$\eta_j(\Omega) = \log \frac{d_j}{d_{j-1}}, \qquad 1 \le j \le K. \tag{4.9}$$

The rigid spectral strain is

$$\Sigma(\Omega) = \|\eta(\Omega)\|_\infty. \tag{4.10}$$

Corollary 4.4. *The positivity test becomes*

$$\|\eta(\Omega)\|_\infty \le \frac{1}{2} \log b. \tag{4.11}$$

The minimum compatible Perron value is

$$b_*(\Omega) = \exp\left(2\|\eta(\Omega)\|_\infty\right) \tag{4.12}$$

$$= \max_{1 \le j \le K} \left\{ \left(\frac{d_j}{d_{j-1}}\right)^2, \left(\frac{d_{j-1}}{d_j}\right)^2 \right\}. \tag{4.13}$$

If some $d_j \le 0$*, set* $b_*(\Omega) = +\infty$*.*

Remark 4.5. If $b > b_*(\Omega)$, the rigid Leslie coefficients are strictly positive. If $b = b_*(\Omega) < \infty$, at least one paired coefficient vanishes. The map b_* is generally nonsmooth where several coordinates attain the maximum in (**??**).

Proposition 4.6. *The rigid root-of-unity comb*

$$\omega_k = \frac{2\pi k}{2K+1}, \qquad 1 \le k \le K, \tag{4.14}$$

has $d_j = 1$ *for all* j*, and hence* $\Sigma(\Omega) = 0$*. Conversely,* $\Sigma(\Omega) = 0$ *implies the same multiset of phases, up to ordering and conjugation.*

Proof. For (**??**),

$$S_\Omega(w) = \frac{w^{2K+1} - 1}{w - 1} = 1 + w + \cdots + w^{2K}. \tag{4.15}$$

Conversely, $\eta = 0$ and $d_0 = 1$ imply $d_0 = \cdots = d_{2K} = 1$. The roots of (**??**) are the nontrivial $(2K+1)$-st roots of unity. □

5. Memory tails and the universal critical region

Normalize the coefficients by

$$x_\ell = \frac{a_\ell}{b^\ell}, \qquad X_q = \sum_{\ell=q}^{m} x_\ell. \tag{5.1}$$

Proposition 5.1. *The tails telescope to*

$$X_{j+1} = \frac{d_j}{r^j}, \qquad 0 \le j \le 2K. \tag{5.2}$$

Consequently,

$$X_{m-j} = b^{j-K} X_{j+1}, \qquad 0 \le j \le K, \tag{5.3}$$

$$x_m = X_m = b^{-K}. \tag{5.4}$$

Moreover, positivity is equivalent to

$$\frac{X_j}{b} \le X_{j+1} \le X_j, \qquad 1 \le j \le K. \tag{5.5}$$

Proof. Substitute (**??**) into (**??**); consecutive terms telescope. Equations (**??**)–(**??**) follow from (**??**), and (**??**) is (**??**) rewritten using (**??**). □

The reciprocal polynomial admits a Chebyshev reduction. Define

$$C_n(y) = 2T_n(y/2), \tag{5.6}$$

where T_n is the Chebyshev polynomial of the first kind, and set

$$H_\eta(y) = C_K(y) + d_1 C_{K-1}(y) + \cdots + d_{K-1} C_1(y) + d_K. \tag{5.7}$$

Then

$$S_\Omega(w) = w^K H_\eta(w + w^{-1}). \tag{5.8}$$

Definition 5.2. Let

$$\mathcal{R}_K = \left\{ \eta \in \mathbb{R}^K : H_\eta \text{ has } K \text{ simple roots in } (-2, 2) \right\}, \tag{5.9}$$

where $d_j = \exp(\eta_1 + \cdots + \eta_j)$.

Proposition 5.3. *In η-coordinates, the admissible rigid critical region is*

$$\mathcal{A}^\eta_{b,K} = \mathcal{R}_K \cap \left[-\frac{1}{2}\log b, \frac{1}{2}\log b \right]^K. \tag{5.10}$$

Thus $\mathcal{R}_K$ encodes the unit-circle root geometry, while the box encodes Leslie positivity.

Remark 5.4. Within the positive-d coordinates, a path can leave (**??**) either through a box face or through the boundary of $\mathcal{R}_K$. The latter occurs when roots collide or reach an endpoint. Since

$$H_\eta(2) = d_K + 2\sum_{j=0}^{K-1} d_j > 0, \tag{5.11}$$

a root cannot reach $+2$. Candidate spectral boundary equations are therefore

$$\operatorname{Disc} H_\eta = 0 \qquad \text{or} \qquad H_\eta(-2) = 0. \tag{5.12}$$

This describes the regular boundary mechanisms; intersections and higher degeneracies form singular strata.

6. Jacobian of the phase-to-strain map

Assume distinct phases $0 < \omega_k < \pi$. Let the phases be ordered to remove permutation ambiguity.

Proposition 6.1. *The coefficient Jacobian satisfies*

$$\left|\det \frac{\partial(d_1, \dots, d_K)}{\partial(\omega_1, \dots, \omega_K)}\right| = 2^{K(K+1)/2} \prod_{k=1}^{K} \sin \omega_k \prod_{i<j} |\cos \omega_i - \cos \omega_j|. \tag{6.1}$$

Consequently,

$$\left|\det \frac{\partial \eta}{\partial \Omega}\right| = \frac{2^{K(K+1)/2} \prod_{k} \sin \omega_k \prod_{i<j} |\cos \omega_i - \cos \omega_j|}{\prod_{j=1}^{K} d_j}. \tag{6.2}$$

Proof. The first K reciprocal coefficients determine the monic polynomial with roots $2\cos\omega_k$ in the Chebyshev basis. The root-to-coefficient Jacobian is a Vandermonde product, the change $2\cos\omega_k$ contributes $2\sin\omega_k$ in absolute value, and the triangular changes between monomial, Chebyshev, and reciprocal coefficient coordinates give the stated power of 2. The map $d \mapsto \eta$ is triangular with determinant $(d_1 \cdots d_K)^{-1}$. □

Corollary 6.2. *The same determinant can be written as*

$$\left|\det \frac{\partial \eta}{\partial \Omega}\right| = \frac{\sqrt{|\mathrm{Disc}\, H_\eta \, H_\eta(2) H_\eta(-2)|}}{d_1 d_2 \cdots d_K}. \tag{6.3}$$

Hence the phase-to-strain map is locally invertible away from phase collisions and the endpoints 0 *and* π*.*

Remark 6.3. At regular points, a moving rigid Leslie model can be transported using K variables η, plus b when the Perron value is allowed to vary. The full coefficient vector is reconstructed from (**??**), (**??**), and (**??**).

7. Exact viability conditions in strain coordinates

For fixed b, the box criterion (**??**) gives the exact first-order viability conditions. On an upper active face,

$$\eta_j = \frac{1}{2}\log b \quad \Longrightarrow \quad \dot\eta_j \le 0, \tag{7.1}$$

and on a lower active face,

$$\eta_j = -\frac{1}{2}\log b \quad \Longrightarrow \quad \dot\eta_j \ge 0. \tag{7.2}$$

Since $\dot\eta = D_\Omega \eta \, \dot\Omega$, both cases are summarized by

$$\mathrm{sgn}(\eta_j) \sum_{k=1}^{K} \frac{\partial \eta_j}{\partial \omega_k} \dot\omega_k \le 0 \tag{7.3}$$

on an active face.

If $b = b(t)$ varies and

$$\eta_j = s_j \frac{1}{2}\log b, \qquad s_j \in \{-1, 1\}, \tag{7.4}$$

then viability requires

$$s_j \dot\eta_j \le \frac{1}{2}\frac{\dot b}{b}. \tag{7.5}$$

For the minimum-Perron choice $b = b_*(\Omega)$ and a unique active index j_*,

$$\frac{\dot{b}_*}{b_*} = 2\,\mathrm{sgn}(\eta_{j_*})\dot{\eta}_{j_*}. \tag{7.6}$$

Remark 7.1. When several coordinates attain $\|\eta\|_\infty$, b_* is not differentiable in general. Directional derivatives or a differential inclusion must replace (**??**).

On a fixed upper face with tangential first-order motion, the second-order viability condition is

$$\nabla\eta_j \cdot \ddot{\Omega} + \dot{\Omega}^T D^2\eta_j \dot{\Omega} \le 0. \tag{7.7}$$

The inequality reverses on a lower face.

8. Exact modal transport and internally generated amplitudes

Assume that every rigid polynomial has distinct roots. For a root λ, define the right eigenvector

$$v(\lambda) = (\lambda^{m-1}, \lambda^{m-2}, \dots, \lambda, 1)^T. \tag{8.1}$$

Let

$$V_j = [v(\lambda_{j,1}) \ \cdots \ v(\lambda_{j,m})], \qquad \Lambda_j = \mathrm{diag}(\lambda_{j,1}, \dots, \lambda_{j,m}). \tag{8.2}$$

If L_j acts for h_j steps and $y_j = V_j c_j$, then

$$y_{j+1} = V_j \Lambda_j^{h_j} c_j, \tag{8.3}$$

$$c_{j+1} = T_j \Lambda_j^{h_j} c_j, \qquad T_j = V_{j+1}^{-1} V_j. \tag{8.4}$$

Proposition 8.1. *The entries of the basis-change matrix are*

$$(T_j)_{pq} = \prod_{\substack{s=1 \\ s\neq p}}^{m} \frac{\lambda_{j,q} - \lambda_{j+1,s}}{\lambda_{j+1,p} - \lambda_{j+1,s}}. \tag{8.5}$$

Proof. The p-th coordinate in the new Vandermonde basis is the cardinal Lagrange polynomial for the nodes $\lambda_{j+1,s}$, evaluated at the old node $\lambda_{j,q}$. Inverse Vandermonde matrices and Lagrange interpolation are equivalent descriptions of this change of basis. □

For a small root displacement

$$\lambda_{j+1,q} = \lambda_{j,q} + \delta\lambda_{j,q}, \tag{8.6}$$

one obtains, for $p \neq q$,

$$(T_j)_{pq} = -\delta\lambda_{j,q} \frac{P_j'(\lambda_{j,q})}{(\lambda_{j,q} - \lambda_{j,p}) P_j'(\lambda_{j,p})} + O(\|\delta\lambda_j\|^2). \tag{8.7}$$

Thus modal mixing becomes large near collisions, consistently with the vanishing of the discriminant.

Proposition 8.2. *The exact modal volume identity is*

$$\det T_j = \frac{\det V_j}{\det V_{j+1}}, \tag{8.8}$$

$$|\det T_j|^2 = \frac{|\operatorname{Disc} P_j|}{|\operatorname{Disc} P_{j+1}|}. \tag{8.9}$$

Proof. The first identity follows from $T_j = V_{j+1}^{-1} V_j$. The square of the absolute Vandermonde determinant equals the absolute discriminant of a monic polynomial with simple roots. □

Iterating (**??**) gives

$$c_J = T_{J-1}\Lambda_{J-1}^{h_{J-1}} \cdots T_0 \Lambda_0^{h_0} c_0, \tag{8.10}$$

and hence

$$y_J = V_J T_{J-1}\Lambda_{J-1}^{h_{J-1}} \cdots T_0 \Lambda_0^{h_0} c_0. \tag{8.11}$$

For the first-coordinate observable,

$$U_J = e_1^T V_J T_{J-1}\Lambda_{J-1}^{h_{J-1}} \cdots T_0 \Lambda_0^{h_0} c_0. \tag{8.12}$$

Warning 8.3. Equations (**??**)–(**??**) eliminate independent amplitude fitting only under the homogeneous state-handoff model (**??**). If each data window is reinitialized, forced, normalized independently, or connected through an additional observation operator, new parameters re-enter and must be counted explicitly.

9. Canonical boundary realization and the remaining law

Whenever $b_*(\Omega) < \infty$, define the canonical minimum-Perron realization

$$L_*(\Omega) = L\big(a(b_*(\Omega), \Omega)\big). \tag{9.1}$$

This realization lies on the boundary of the nonnegative cone. At regular points with unique active strain coordinate, the maps

$$\Omega \longleftrightarrow \eta \longmapsto b_*(\Omega) \longmapsto L_*(\Omega) \tag{9.2}$$

are locally determined, with the first arrow locally invertible.

Remark 9.1. The construction (**??**) is canonical but only piecewise smooth because of the maximum norm in b_*. It is also numerically sensitive near the spectral boundary (**??**).

The static coordinate chain (**??**) does not by itself determine how a future window is reached. The next sections derive an exact overlap law directly from the scalar sequence. For fixed b, the special reciprocal structure turns the admissible coefficient differences into a linear problem. At the first generic stride admitting motion, the entire update is reduced to one scalar innovation.

10. The fixed-Perron reciprocal affine hull

The critical-circle condition has a useful linear precursor. Fix $b > 1$, put $m = 2K + 1$, and for a coefficient vector $a = (a_1, \dots, a_m)$ define the synthetic-division coefficients

$$q_0(a; b) = 1, \tag{10.1}$$

$$q_j(a; b) = b^j - \sum_{\ell=1}^{j} a_\ell b^{j-\ell}, \qquad 1 \le j \le 2K. \tag{10.2}$$

Then

$$P(z; a) = (z - b)\left(z^{2K} + q_1 z^{2K-1} + \cdots + q_{2K}\right) + P(b; a). \tag{10.3}$$

Definition 10.1. The fixed-b reciprocal–Perron affine hull is the set

$$\mathcal{C}_{b,K}^{\mathrm{rec}} = \left\{ a \in \mathbb{R}^m : P(b; a) = 0, \quad q_{2K-j}(a; b) = b^{K-j} q_j(a; b),\ 0 \le j \le K - 1 \right\}. \tag{10.4}$$

Proposition 10.2. *The set* (**??**) *is an affine subspace of dimension K. Equivalently, there is a rank-$(K + 1)$ matrix R_b and a vector c_b such that*

$$\mathcal{C}_{b,K}^{\mathrm{rec}} = \{a : R_b a = c_b\}. \tag{10.5}$$

One explicit choice of the $K+1$ affine rows is

$$\sum_{\ell=1}^{m} b^{m-\ell} a_\ell = b^m, \tag{10.6}$$

$$\sum_{\ell=1}^{2K-j} b^{2K-j-\ell} a_\ell - \sum_{\ell=1}^{j} b^{K-\ell} a_\ell = b^{2K-j} - b^K, \qquad 0 \le j \le K-1. \tag{10.7}$$

Every rigid critical Leslie coefficient vector with Perron value b belongs to this affine subspace.

Proof. All equations in (**??**) are affine linear in a. The K quantities $q_1, \dots, q_K$ may be chosen freely. The reciprocal identities determine $q_{K+1}, \dots, q_{2K}$, after which

$$a_j = b q_{j-1} - q_j, \qquad 1 \le j \le 2K, \tag{10.8}$$

and $P(b; a) = 0$ determines

$$a_m = b q_{2K} = b^{K+1}. \tag{10.9}$$

Hence the affine dimension is K. For a rigid critical spectrum, $q_j = r^j d_j$ with $r = \sqrt{b}$, and the palindromy $d_{2K-j} = d_j$ gives exactly the reciprocal equations in (**??**). □

Remark 10.3. Membership in $\mathcal{C}^{\mathrm{rec}}_{b,K}$ is necessary but not sufficient for all non-Perron roots to lie on $|z| = \sqrt{b}$. Put

$$d_j = \frac{q_j}{b^{j/2}}. \tag{10.10}$$

The reciprocal identities give $d_{2K-j} = d_j$. Full criticality additionally requires the associated polynomial H in (**??**) to have all K roots in $[-2, 2]$, together with the Leslie nonnegativity inequalities.

11. Overlapping windows and exact mobility rigidity

Let (u_t) be any scalar sequence. In the intended arithmetic application one may take $u_t = M(B_t)$ or a predeclared normalization of those values. Define the recurrence row

$$h_t = (u_{t-1}, u_{t-2}, \dots, u_{t-m}) \in \mathbb{R}^m. \tag{11.1}$$

The order-m recurrence window beginning at n consists of the m equations

$$h_t a^{(n)} = u_t, \qquad n+m \le t \le n+2m-1. \tag{11.2}$$

When the corresponding square data matrix is nonsingular, these equations determine $a^{(n)}$ uniquely.

Let a second window begin at $n+s$, with $1 \le s \le m$. The two recurrence windows share exactly $m-s$ equations. Let $O_{n,s}$ be the matrix whose rows are the common h_t:

$$O_{n,s} = \begin{pmatrix} h_{n+s+m} \\ \vdots \\ h_{n+2m-1} \end{pmatrix} \in \mathbb{R}^{(m-s)\times m}, \tag{11.3}$$

with the empty-matrix convention when $s = m$.

Theorem 11.1 (Exact overlap constraint)**.** *Assume that $a^{(n)}$ and $a^{(n+s)}$ both fit their respective windows and both belong to $\mathcal{C}^{\mathrm{rec}}_{b,K}$ for the same fixed b. Then*

$$\delta a_{n,s} := a^{(n+s)} - a^{(n)} \in \ker O_{n,s} \cap \ker R_b. \tag{11.4}$$

Equivalently, with

$$Z_{n,s} = \begin{pmatrix} O_{n,s} \\ R_b \end{pmatrix}, \tag{11.5}$$

one has

$$Z_{n,s}\delta a_{n,s} = 0. \tag{11.6}$$

Proof. Every common recurrence equation is satisfied by both coefficient vectors, so subtraction gives $O_{n,s}\delta a_{n,s} = 0$. Since both vectors satisfy the same affine equations $R_b a = c_b$, subtraction also gives $R_b \delta a_{n,s} = 0$. □

Corollary 11.2 (Generic mobility threshold)**.** *If the rows of $O_{n,s}$ and R_b are in generic relative position, then*

$$\dim\ker Z_{n,s} = \max\{0, s-K-1\}. \tag{11.7}$$

Hence:

$$s \le K+1 \implies \textit{the old coefficient vector is locally isolated}, \tag{11.8}$$

$$s = K+2 \implies \textit{one generic coefficient direction survives}, \tag{11.9}$$

$$s > K+2 \implies s-K-1 \textit{ generic directions survive}. \tag{11.10}$$

Proof. The common-equation matrix contributes $m-s$ generic independent rows and R_b contributes $K+1$ independent rows. Since $m = 2K+1$, the generic nullity is

$$m - \min\{m, m-s+K+1\} = \max\{0, s-K-1\}. \tag{11.11}$$

□

Definition 11.3. For fixed b, define the critical stride

$$s_* = K+2. \tag{11.12}$$

It is the first stride at which a nonzero fixed-b reciprocal–Perron motion exists generically.

Remark 11.4. For $s \le K+1$, nontrivial motion is possible only when $Z_{n,s}$ loses rank. Thus exceptional motion is detected by the vanishing of maximal minors of $Z_{n,s}$. In particular, any single Hankel-type minor obtained from the fixed identity $a_m = b^{K+1}$ is only one necessary degeneracy condition, not a sufficient condition for full critical motion.

Remark 11.5. If b is allowed to vary, the regular rigid critical manifold has dimension $K+1$. A transversality count gives local intersection dimension

$$\max\{0, s-K\}, \tag{11.13}$$

so the first generic one-dimensional mobility occurs at $s = K+1$. Unlike the fixed-b result, this statement is local and nonlinear because coefficient differences for distinct Perron values do not lie in one fixed affine subspace.

12. Scalar innovation at the critical stride

Set $s = s_* = K+2$ and assume

$$\operatorname{rank} Z_{n,s_*} = m-1. \tag{12.1}$$

Choose a nonzero vector v_n spanning the one-dimensional nullspace,

$$\ker Z_{n,s_*} = \operatorname{span}\{v_n\}. \tag{12.2}$$

Theorem 12.1 (Scalarization of the window update)**.** *Under* (**??**)*, every next fixed-b reciprocal–Perron coefficient vector that agrees with the old window on all common equations has the form*

$$a^{(n+s_*)} = a^{(n)} + \tau_n v_n \tag{12.3}$$

for a unique scalar τ_n.

Proof. The difference belongs to the one-dimensional nullspace in (**??**). □

The next window contributes $s_* = K + 2$ genuinely new recurrence equations. The first occurs at

$$t_0 = n + 2m. \tag{12.4}$$

Its recurrence row h_{t_0} uses only values already known before u_{t_0} is observed.

Theorem 12.2 (One innovation sample determines the next Leslie)**.** *Assume*

$$h_{t_0} v_n \neq 0. \tag{12.5}$$

Then the first new scalar value determines the update parameter exactly:

$$\boxed{\tau_n = \frac{u_{t_0} - h_{t_0} a^{(n)}}{h_{t_0} v_n}.} \tag{12.6}$$

Consequently the entire next coefficient vector is fixed by (**??**)*.*

Proof. Substitute (**??**) into the recurrence equation at t_0 and solve the resulting scalar equation for τ_n. □

Corollary 12.3 (Causal $K+1$-value forecast)**.** *After observing only u_{t_0}, define $a^+ = a^{(n)} + \tau_n v_n$ using* (**??**)*. Then recursively define*

$$\widehat{u}_t = \sum_{\ell=1}^{m} a_\ell^+ \widehat{u}_{t-\ell}, \qquad t_0 + 1 \le t \le t_0 + K + 1, \tag{12.7}$$

using the observed historical values for indices below t_0 and earlier forecasts above t_0. If the next window is exactly generated by a fixed-b rigid critical Leslie coefficient vector, then

$$\widehat{u}_t = u_t, \qquad t_0 + 1 \le t \le t_0 + K + 1. \tag{12.8}$$

Proof. The first new observation determines the exact next coefficient vector. The remaining $K+1$ new equations of that window use the same coefficient vector. Induction on t in (**??**) gives the result. □

Proposition 12.4 (Rank-one block diagnostic)**.** *Let G_n contain the $K+2$ new recurrence rows of the next window, evaluated from the observed sequence, and let y_n^+ be their right-hand side. Define*

$$e_n = y_n^+ - G_n a^{(n)}, \qquad g_n = G_n v_n. \tag{12.9}$$

Then exact critical-stride transport implies

$$e_n = \tau_n g_n. \tag{12.10}$$

Equivalently,

$$(e_n)_i (g_n)_j - (e_n)_j (g_n)_i = 0 \qquad \text{for all } i, j. \tag{12.11}$$

Thus a block of $K+2$ new equations carries only one scalar innovation after the old state and overlap direction are fixed.

13. Admissible innovation interval and boundary locking

The scalar parameter in (**??**) cannot take arbitrary values if nonnegativity and the critical-circle root condition are retained.

Definition 13.1. For a current nonnegative coefficient vector a and direction v, define the interval of coefficient positivity

$$I_{\mathrm{pos}}(a, v) = \left[\max_{v_\ell > 0} \left(-\frac{a_\ell}{v_\ell} \right),\ \min_{v_\ell < 0} \left(-\frac{a_\ell}{v_\ell} \right) \right], \tag{13.1}$$

with the usual $-\infty$ and $+\infty$ conventions when one of the indexing sets is empty.

For $a(\tau) = a + \tau v$, define $q_j(\tau)$ by (**??**), $d_j(\tau) = q_j(\tau)/b^{j/2}$, and the corresponding reciprocal reduction H_τ. Let I_{spec} be the connected component containing 0 of the set of τ for which H_τ has K simple roots in $(-2, 2)$. At regular points, an endpoint of this component can occur only when

$$\operatorname{Disc} H_\tau = 0 \qquad \text{or} \qquad H_\tau(-2) = 0. \tag{13.2}$$

Define the full local innovation interval

$$I_{\mathrm{adm}} = I_{\mathrm{pos}} \cap I_{\mathrm{spec}}. \tag{13.3}$$

Corollary 13.2 (Pre-observation admissible range)**.** *Before u_{t_0} is observed, the moving rigid Leslie hypothesis requires*

$$\boxed{u_{t_0} \in h_{t_0} a^{(n)} + (h_{t_0} v_n) I_{\mathrm{adm}}.} \tag{13.4}$$

A value outside this interval falsifies the local fixed-b critical transition before any later value of the new block is used.

Proposition 13.3 (Boundary locking criterion)**.** *Let*

$$\mathcal{Z}(a) = \{\ell : a_\ell = 0\} \tag{13.5}$$

be the active coefficient faces. A nonzero one-sided coefficient motion $a+\tau v$ preserving nonnegativity exists locally only if the numbers v_ℓ, $\ell \in \mathcal{Z}(a)$, do not contain both positive and negative values. If both signs occur, then $\tau = 0$ is the only sufficiently small feasible update.

Proof. For each active coordinate the inequality $a_\ell + \tau v_\ell \geq 0$ reduces to $\tau v_\ell \geq 0$. A common nonzero sign of τ exists exactly when the active v_ℓ do not impose conflicting signs. □

Remark 13.4. The canonical minimum-Perron realization $L_*(\Omega)$ always has at least one active coefficient face. It is therefore dynamically more restrictive than an interior choice $b > b_*(\Omega)$. A canonical boundary trajectory may lock whenever the critical-stride direction intersects active faces with incompatible signs.

14. Exact scalar-driven strain transport

For the update $a^+ = a + \tau v$, define

$$\psi_j(v; b) = \sum_{\ell=1}^{j} v_\ell b^{j-\ell}, \qquad \psi_0 = 0. \tag{14.1}$$

Equation (**??**) gives

$$q_j(a + \tau v; b) = q_j(a; b) - \tau \psi_j(v; b). \tag{14.2}$$

Theorem 14.1 (Exact strain update)**.** *Whenever all logarithms are defined,*

$$\boxed{\eta_j^+ - \eta_j = \log\left(1 - \tau \frac{\psi_j}{q_j}\right) - \log\left(1 - \tau \frac{\psi_{j-1}}{q_{j-1}}\right), \qquad 1 \leq j \leq K.} \tag{14.3}$$

Hence, at the critical stride, all K strain coordinates are driven by the same scalar innovation τ.

Proof. Use $d_j = q_j/b^{j/2}$ in the definition $\eta_j = \log(d_j/d_{j-1})$ and substitute (**??**). □

Differentiating with respect to the scalar innovation gives the exact line-flow equation

$$\frac{d\eta_j}{d\tau} = -\frac{\psi_j}{q_j - \tau\psi_j} + \frac{\psi_{j-1}}{q_{j-1} - \tau\psi_{j-1}}. \tag{14.4}$$

For small τ,

$$\eta_j^+ - \eta_j = -\tau\left(\frac{\psi_j}{q_j} - \frac{\psi_{j-1}}{q_{j-1}}\right) + O(\tau^2). \tag{14.5}$$

Thus the K-dimensional strain increment is generically confined to one first-order direction at the critical stride.

15. Frequency-free determinant identities

The reciprocal–Perron test can be imposed directly on a data window without first estimating any phases. Let

$$\mathsf{A}_n = \begin{pmatrix} h_{n+m} \\ \vdots \\ h_{n+2m-1} \end{pmatrix}, \qquad \mathsf{g}_n = \begin{pmatrix} u_{n+m} \\ \vdots \\ u_{n+2m-1} \end{pmatrix}. \tag{15.1}$$

Write the affine reciprocal constraints in (**??**) row by row as

$$\rho_\alpha(b)^T a = \kappa_\alpha(b), \qquad 0 \le \alpha \le K. \tag{15.2}$$

For each row define

$$\Delta_{\alpha,n}(b) = \det \begin{pmatrix} \mathsf{A}_n & \mathsf{g}_n \\ \rho_\alpha(b)^T & \kappa_\alpha(b) \end{pmatrix}, \tag{15.3}$$

Proposition 15.1. *If* $\det \mathsf{A}_n \neq 0$*, then the unique order-*m *recurrence coefficient vector of the window belongs to* $\mathcal{C}^{\mathrm{rec}}_{b,K}$ *if and only if*

$$\Delta_{\alpha,n}(b) = 0, \qquad 0 \le \alpha \le K. \tag{15.4}$$

Full rigid criticality is obtained by adding the nonnegativity and root-location tests of Sections **??**–**??**.

Proof. Since $a = \mathsf{A}_n^{-1}\mathsf{g}_n$, the Schur-complement identity gives

$$\Delta_{\alpha,n}(b) = \det(\mathsf{A}_n)\left(\kappa_\alpha(b) - \rho_\alpha(b)^T \mathsf{A}_n^{-1}\mathsf{g}_n\right). \tag{15.5}$$

Thus the determinants vanish exactly when all affine reciprocal–Perron constraints hold. □

Remark 15.2. If b is not fixed in advance, the functions $\Delta_{\alpha,n}(b)$ are explicit polynomials in b whose coefficients depend only on the data window. A necessary condition for a rigid critical realization with some $b > 1$ is that these $K+1$ polynomials possess a common real root $b > 1$. This can be tested without frequencies by polynomial gcd or resultant/subresultant elimination, followed by the root-location and positivity conditions.

16. Closed form for one mode pair

For $K = 1$ one has $m = 3$, and the entire construction becomes explicit. Write $\alpha = a_1$. The fixed-b reciprocal–Perron constraints give

$$\boxed{a_1 = \alpha, \qquad a_2 = b^2 - b - b\alpha, \qquad a_3 = b^2.} \tag{16.1}$$

The quotient polynomial is

$$Q(z) = z^2 + (b - \alpha)z + b. \tag{16.2}$$

Its two roots lie on $|z| = \sqrt{b}$, and the Leslie coefficients are nonnegative, if and only if

$$\boxed{\max\{0, b - 2\sqrt{b}\} \le \alpha \le b - 1.} \tag{16.3}$$

Endpoint equalities include coefficient or spectral degeneracies; strict interior conditions give the corresponding nondegenerate case.

Define

$$X_t(b) = u_{t-1} - bu_{t-2}, \tag{16.4}$$

$$Y_t(b) = u_t - (b^2 - b)u_{t-2} - b^2 u_{t-3}. \tag{16.5}$$

Then the critical recurrence is simply

$$Y_t(b) = \alpha X_t(b). \tag{16.6}$$

Thus, if (X_t, X_{t+1}, X_{t+2}) is not identically zero, three consecutive recurrence equations admit a common fixed-b reciprocal coefficient vector if and only if all pairwise minors vanish:

$$X_i Y_j - X_j Y_i = 0, \qquad t \le i < j \le t+2. \tag{16.7}$$

If $X_t = X_{t+1} = X_{t+2} = 0$, existence instead requires $Y_t = Y_{t+1} = Y_{t+2} = 0$. Full rigid criticality additionally requires the inferred α to satisfy (**??**).

For $K = 1$, the critical stride is $s_* = 3$ and the fixed-b affine direction is

$$v = (1, -b, 0)^T. \tag{16.8}$$

Hence

$$\alpha^+ = \alpha + \tau, \qquad a_2^+ = a_2 - b\tau, \qquad a_3^+ = b^2. \tag{16.9}$$

The first new recurrence equation gives

$$\tau = \frac{u_{t_0} - h_{t_0} a}{X_{t_0}(b)}, \tag{16.10}$$

provided $X_{t_0}(b) \neq 0$, and the next two values are then fixed recursively. The exact admissible scalar interval is

$$\boxed{\max\{0, b - 2\sqrt{b}\} - \alpha \le \tau \le b - 1 - \alpha.} \tag{16.11}$$

This gives a completely frequency-free three-step falsification and two-value forecasting rule for the one-pair model.

17. Irreducible scalar forcing and the endpoint of the Leslie geometry

Proposition 17.1. *At a regular interior fixed-b point and the critical stride $s_* = K + 2$, the Leslie geometry cannot determine τ_n from the old window and the common equations alone. In a neighborhood of $\tau = 0$, distinct values of τ produce distinct reciprocal–Perron coefficient vectors satisfying all those old and common constraints. Full criticality persists for a nontrivial interval whenever the current point lies in the interior of the fixed-b critical region.*

Proof. The rank assumption (**??**) leaves exactly the affine line (**??**). Within the reciprocal affine hull, coefficient positivity and the condition that the reduced polynomial have simple roots in $(-2, 2)$ are open. Hence sufficiently small values of τ remain admissible. □

Remark 17.2. This identifies the exact limit of the fixed-b rigid moving-Leslie mechanism. The rigid geometry compresses a block of $K + 2$ genuinely new recurrence equations to one scalar innovation and then transports all K strain coordinates through (**??**). It does not supply a law for that scalar innovation. A zero-latency forecast of the first new value therefore requires an additional arithmetic statement about the sequence τ_n. Any such law is external to Leslie geometry and is the remaining place where genuinely Möbius-specific information would have to enter.

18. Exact first-order spectral transport

Let $a(t)$ be a differentiable coefficient path and define

$$P(z,t) = z^m - \sum_{\ell=1}^{m} a_\ell(t) z^{m-\ell}. \tag{18.1}$$

Let $\lambda_k(t)$ be a simple root, so that

$$P(\lambda_k(t),t) = 0, \qquad P_z(\lambda_k(t),t) \neq 0. \tag{18.2}$$

Differentiability of simple eigenvalue branches is standard in matrix perturbation theory [**?**].

Proposition 18.1. *Under* (**??**),

$$\dot\lambda_k = \frac{\displaystyle\sum_{\ell=1}^{m} \dot a_\ell \lambda_k^{m-\ell}}{P_z(\lambda_k,t)}. \tag{18.3}$$

Proof. Differentiating $P(\lambda_k(t),t) = 0$ gives

$$P_z(\lambda_k,t)\dot\lambda_k + P_t(\lambda_k,t) = 0. \tag{18.4}$$

Since

$$P_t(z,t) = -\sum_{\ell=1}^{m} \dot a_\ell z^{m-\ell}, \tag{18.5}$$

substitution into (**??**) yields (**??**). □

Suppose now that

$$\lambda_k(t) = \sqrt{b(t)}\, e^{i\omega_k(t)}. \tag{18.6}$$

Then

$$\frac{\dot\lambda_k}{\lambda_k} = \frac{\dot b}{2b} + i\dot\omega_k. \tag{18.7}$$

Define

$$J_{k\ell}(L) = \frac{\lambda_k^{m-\ell-1}}{P_z(\lambda_k)}. \tag{18.8}$$

Corollary 18.2. *For every simple prescribed branch,*

$$\dot\omega_k = \Im\left(\sum_{\ell=1}^{m} J_{k\ell}(L)\dot a_\ell\right), \tag{18.9}$$

$$\frac{\dot b}{2b} = \Re\left(\sum_{\ell=1}^{m} J_{k\ell}(L)\dot a_\ell\right). \tag{18.10}$$

Warning 18.3. Equation (**??**) must give the same scalar $\dot b/(2b)$ for every k. Thus the imaginary-part formula alone is not a complete admissibility test. The real-part compatibility equations are additional local constraints.

19. The rigid critical manifold

Let

$$\mathcal{M}_K = \left\{ a(b,\Omega) : b > 1,\ \Omega \in \mathbb{T}^K,\ a(b,\Omega) \geq 0 \right\} \subseteq \mathbb{R}^{2K+1}. \tag{19.1}$$

Proposition 19.1. *At a point where the coefficient map $(b,\Omega) \mapsto a(b,\Omega)$ has rank $K+1$ and the prescribed roots are distinct, $\mathcal{M}_K$ is locally a smooth manifold of dimension $K+1$ and codimension K in $\mathbb{R}^{2K+1}$.*

Proof. The parameter space has dimension $K+1$. Under the stated full-rank hypothesis, the constant-rank theorem gives a local image manifold of dimension $K+1$. The ambient dimension is $2K+1$, so the codimension is K. □

Consequently, locally there are K independent constraints

$$F_r(a) = 0, \qquad 1 \leq r \leq K, \tag{19.2}$$

and every differentiable path in the regular part satisfies

$$\nabla F_r(a) \cdot \dot{a} = 0, \qquad 1 \leq r \leq K. \tag{19.3}$$

Warning 19.2. The dimension statement is generic, not universal. Repeated roots, collisions of phases, loss of rank, and intersections with several active coefficient faces can create singular or lower-dimensional strata.

20. Positivity and the corrected spectral velocity cone

For $a \in \mathbb{R}^m_+$, the tangent cone of the nonnegative orthant is

$$T_a\mathbb{R}^m_+ = \{ v \in \mathbb{R}^m : v_\ell \geq 0 \text{ whenever } a_\ell = 0 \}. \tag{20.1}$$

Tangent cones generalize tangent spaces at nonsmooth boundary points of convex sets [**?**].

The velocity must satisfy three types of conditions simultaneously:

(1) orthant feasibility, $v \in T_a\mathbb{R}^m_+$;
(2) tangency to the rigid critical manifold, $DF(a)v = 0$;
(3) common radial motion across all prescribed root pairs.

Define the common-radial constraint operator by

$$\mathcal{R}_L v = \left(\Re(J_2(L)v) - \Re(J_1(L)v), \ldots, \Re(J_K(L)v) - \Re(J_1(L)v) \right)^T. \tag{20.2}$$

At a regular point, define the admissible coefficient-velocity cone

$$\mathcal{K}(L) = T_a\mathbb{R}^m_+ \cap \ker DF(a) \cap \ker \mathcal{R}_L. \tag{20.3}$$

Definition 20.1. The rigid positive spectral velocity cone is

$$\mathcal{V}(L) = \{ \Im J(L)v : v \in \mathcal{K}(L) \}. \tag{20.4}$$

Proposition 20.2. *Every differentiable rigid positive Leslie path with simple prescribed roots satisfies*

$$\dot{\Omega}(t) \in \mathcal{V}(L(t)). \tag{20.5}$$

Proof. The coefficient velocity lies in the orthant tangent cone, is tangent to the rigid critical manifold, and satisfies the common-radial identities. Equation (**??**) then places its phase velocity in the image (**??**). □

Remark 20.3. Condition (**??**) is first-order and necessary. At singular points or for finite window increments it is not sufficient for the existence of a global path.

21. Second-order transport

Proposition 21.1. *If $a(t)$ is twice differentiable and (??) holds, then*

$$\ddot{\lambda}_k = -\frac{P_{zz}\dot{\lambda}_k^2 + 2P_{zt}\dot{\lambda}_k + P_{tt}}{P_z}, \tag{21.1}$$

where all polynomial derivatives are evaluated at $(\lambda_k(t), t)$ and

$$P_{tt}(z,t) = -\sum_{\ell=1}^{m} \ddot{a}_\ell z^{m-\ell}. \tag{21.2}$$

Proof. Differentiate (??) once more and collect the chain-rule terms. Division by $P_z \neq 0$ gives (??). □

For the parameterization (??),

$$\frac{\ddot{\lambda}_k}{\lambda_k} = \frac{\ddot{b}}{2b} - \frac{\dot{b}^2}{4b^2} - \dot{\omega}_k^2 + i\left(\ddot{\omega}_k + \frac{\dot{b}}{b}\dot{\omega}_k\right). \tag{21.3}$$

Thus a smooth Leslie path constrains not only Ω and $\dot{\Omega}$, but also the angular acceleration after the radial terms are removed.

Definition 21.2. At a regular path point, the residual obtained by comparing the observed $\ddot{\Omega}$ with the imaginary part implied by (??) is called the local Leslie spectral-curvature residual.

Warning 21.3. A numerical curvature residual depends on the chosen interpolation of discrete windows and on the time scale assigned to the window index. It is not intrinsic until those choices are fixed before testing.

22. Discrete windows and finite increments

The observations are discrete, not a differentiable curve. Let

$$\Delta a_j = a_{j+1} - a_j, \qquad \Delta\Omega_j = \operatorname{unwrap}(\Omega_{j+1} - \Omega_j). \tag{22.1}$$

The differential model suggests the residual

$$r_j = \Delta\Omega_j - \Im J(L_j)\Delta a_j. \tag{22.2}$$

Warning 22.1. Equation (??) is only a first-order approximation. A valid finite-step test should also check that the segment or reconstructed path remains in the admissible set and should report an error scale of order $\|\Delta a_j\|^2$ away from singularities.

A stronger discrete feasibility test is to solve for a path $a_j(s)$, $0 \le s \le 1$, satisfying

$$a_j(0) = a_j, \qquad a_j(1) = a_{j+1}, \qquad a_j(s) \in \mathcal{M}_K \cap \mathbb{R}_+^m. \tag{22.3}$$

Existence of (??) is stronger than endpoint feasibility.

23. A genuine forecasting protocol

Experiment 23.1. Choose before examining the test windows:

(1) a fixed base b or a predeclared rule for b_j;
(2) a fixed number K of mode pairs and hence $m = 2K + 1$;
(3) consecutive, nonoverlapping or explicitly overlapping logarithmic windows;
(4) a frequency estimator and a deterministic mode-matching rule;
(5) a coefficient-transport law trained only on past windows;
(6) a rule for handling windows outside $\mathcal{A}_{b,K}$;
(7) a horizon and evaluation metric.

For each training endpoint J, infer $L_0, \dots, L_J$, predict $\widehat{a}_{J+1}$ without using W_{J+1}, project to the admissible rigid set if that projection was predeclared, compute $\widehat{\Omega}_{J+1}$ from the roots, and only then evaluate the prediction on W_{J+1}.

The information flow must be

$$M|_{W_0,\dots,W_J} \longrightarrow L_0, \dots, L_J \longrightarrow \widehat{L}_{J+1} \longrightarrow \widehat{\Omega}_{J+1} \longrightarrow \widehat{M}|_{W_{J+1}}. \tag{23.1}$$

Warning 23.2. The identities of Sections **??**–**??** do not themselves specify a forecasting law for $\dot{a}$ or $\ddot{a}$. A predictor requires a predeclared dynamical model for coefficient evolution. Merely estimating L_{J+1} from the future window would return to independent fitting.

24. Critical-stride forecasting protocol

Experiment 24.1. Fix K, $m = 2K + 1$, a Perron value $b > 1$, and the critical stride $s_* = K + 2$ before examining the test region. For each training window:

(1) solve the m recurrence equations (**??**) directly from the past scalar sequence;
(2) reject the window unless the resulting coefficients satisfy the fixed-b reciprocal determinants (**??**), nonnegativity, and the critical-circle root-location test;
(3) construct Z_{n,s_*} using only common equations and R_b;
(4) require $\operatorname{rank} Z_{n,s_*} = m - 1$ and compute its null direction v_n;
(5) compute the pre-observation interval (**??**);
(6) reveal only the first new scalar u_{t_0} and compute τ_n from (**??**);
(7) without recalibration, predict the remaining $K + 1$ new values using (**??**);
(8) record the $K + 1$ forecast errors, the scalar innovation τ_n, the strain update (**??**), and whether $\tau_n \in I_{\mathrm{adm}}$.

The information flow in this strongest fixed-b test is therefore

$$\text{past block} \longrightarrow (a^{(n)}, v_n, I_{\mathrm{adm}}) \longrightarrow u_{t_0} \longrightarrow \tau_n \longrightarrow \widehat{u}_{t_0+1}, \dots, \widehat{u}_{t_0+K+1}, \tag{24.1}$$

where u_{t_0} is the single newly revealed scalar value; no later test value is used to update the coefficients.

Remark 24.2. If the $K + 1$-value forecasts succeed repeatedly, the remaining arithmetic object is the scalar innovation sequence (τ_n). A further predictive law for τ_n would upgrade the one-sample-delayed forecast to a genuine zero-latency block forecast. Leslie geometry alone does not provide such a law by Proposition **??**.

25. Four nested models

Definition 25.1. The comparison should separate:

Model A. free frequencies or unconstrained recurrences fitted independently in every window;
Model B. rigid nonnegative Leslie matrices fitted independently in every window;
Model C. one dynamically transported rigid nonnegative Leslie trajectory, forecast from past windows only;
Model D. one dynamically transported *data-compatible repaired* Leslie trajectory with fixed K, fixed auxiliary degree q, and fixed or predeclared Perron rule.

Model B tests static rigid realizability. Model C tests rigid transport. Model D is the appropriate extension when auxiliary roots are genuinely necessary, but it is admissible only after the repaired overlap law has been verified; independently minimizing an auxiliary completion in each window is not Model D. Models C and D should be compared not only with Models A and B, but also with persistence, autoregressive, low-rank Hankel, phase-randomized, support-preserving, and multiplicative null models.

26. What the previous forty runs do and do not establish

Experiment 26.1. The reported forty runs varied the sampling base, training fraction, and requested number of modes. They established numerical out-of-sample prediction by free frequencies and static positive spectral completion with auxiliary eigenvalues for those forty configurations.

Proposition 26.2. *The forty configurations do not constitute a temporal Leslie trajectory and cannot test* (**??**) *or the forecast protocol* (**??**).

Proof. A temporal test requires consecutively ordered windows under fixed or predeclared modelling choices. The forty runs index alternative parameter configurations rather than successive time windows of one fixed model. □

27. Verified corrections to the proposed formulation

Proposition 27.1. *The first derivative* (**??**)*, the angular and radial decompositions* (**??**)–(**??**)*, and the second derivative* (**??**) *are algebraically correct under the simple-root hypothesis.*

Remark 27.2. An independent finite-difference calculation was performed for a degree-five companion polynomial with randomly chosen coefficient velocities. The maximum discrepancy between the finite-difference root velocity and (**??**) was below 9×10^{-9} for a step of 10^{-7}. This is a numerical consistency check, not a substitute for Proposition **??**.

Remark 27.3. Three corrections are essential:

(1) the boundary relation is the inclusion (**??**), not automatic equality;
(2) the spectral velocity cone must include tangency to the rigid manifold and common radial compatibility, as in (**??**);
(3) the dimension and codimension counts are generic and require a full-rank hypothesis.

28. Numerical obstruction for the rigid $K = 7$ core

The rigid feasibility criterion of Corollary **??** can be evaluated even when the logarithmic coordinates η_j do not exist. For $r = \sqrt{b}$ define

$$\delta_j(b, \Omega) = \min\left\{ r d_{j-1} - d_j, r d_j - d_{j-1} \right\}, \qquad 1 \le j \le K, \tag{28.1}$$

and

$$\Delta_K(b, \Omega) = \min_{1 \le j \le K} \delta_j(b, \Omega). \tag{28.2}$$

Because the factors multiplying the two quantities in (**??**) in the paired Leslie coefficients are positive, one has

$$\Delta_K(b, \Omega) \ge 0 \quad \Longleftrightarrow \quad P_{b,\Omega} \text{ is the characteristic polynomial of a nonnegative rigid Leslie matrix.} \tag{28.3}$$

Unlike $b_*(\Omega)$, the margin (**??**) remains defined when one or more d_j are nonpositive.

Experiment 28.1. A grid of 241 values of b in $[1.01, 1.25]$ was examined with $K = 7$ and the predeclared frequency estimator. The estimator returned all seven frequencies at eight grid points. For every one of these eight evaluable configurations,

$$\Delta_7(b, \Omega) < 0. \tag{28.4}$$

The largest observed margin was

$$\boxed{\Delta_{\max} = -1.228610846762} \qquad \text{at} \qquad \boxed{b = 1.178}. \tag{28.5}$$

Thus none of the eight evaluable critical spectra is the complete spectrum of a nonnegative Leslie matrix of rigid dimension 15.

Seven of the eight configurations had $d_1, \dots, d_7 > 0$ and therefore admitted the logarithmic coordinates η and the finite quantity $b_*(\Omega)$. In all seven cases $b_*(\Omega) > b$. The eighth configuration had all seven frequencies but at least one nonpositive d_j, so b_* was not finite although Δ_7 remained evaluable.

The index attaining the most negative paired margin had the empirical distribution

(28.6)

j_*	2	3	4	5	6
count	1	1	3	1	2.

In particular, $j_* = 1$ did not occur. The dominant obstruction in these eight cases is therefore not the first hemispheric condition alone; it occurs in higher symmetric combinations of the seven phases.

Remark 28.2. Experiment **??** is a statement about the eight grid points at which the full $K = 7$ core was available. The other 233 points are not evaluations of Δ_7 and must not be counted as rigidly feasible or infeasible. Likewise, a reported prefix statistic $\kappa = 0$ only excludes the tested angularly ordered prefixes; it does not exclude every possible subfamily of the seven frequencies.

Corollary 28.3. *For each configuration in Experiment* **??** *for which a nonnegative Leslie realization containing the same prescribed roots exists in a larger dimension, at least one auxiliary root is necessary.*

Proof. Equation (**??**) excludes degree-zero completion, i.e. the rigid characteristic polynomial itself. Any larger realization containing the same roots therefore has a nonconstant complementary factor. □

29. Minimal auxiliary completion as an exact linear feasibility problem

The negative rigid test does not determine how many auxiliary eigenvalues are necessary. That question has an exact finite-dimensional formulation that does not require re-estimating any frequency.

Let

$$C(z) = z^d + c_1 z^{d-1} + \cdots + c_d, \qquad d = 2K + 1, \tag{29.1}$$

be a fixed rigid core polynomial $P_{b,\Omega}$, so $C(b) = 0$. For $q \geq 0$ let

$$R_q(z) = z^q + r_1 z^{q-1} + \cdots + r_q \tag{29.2}$$

be a real monic polynomial and write

$$C(z)R_q(z) = z^{d+q} + \sum_{k=1}^{d+q} p_k z^{d+q-k}. \tag{29.3}$$

Definition 29.1. The auxiliary completion index of the core C is

$$\nu(C) = \min\left\{q \geq 0 : \exists R_q \text{ monic real with } p_k \leq 0 \text{ for every } k \geq 1\right\}, \tag{29.4}$$

with $\nu(C) = +\infty$ if no such polynomial exists. For $C = P_{b,\Omega}$ we also write $\nu_b(\Omega) = \nu(C)$.

Proposition 29.2. *A degree-q auxiliary completion in Definition* **??** *exists if and only if there is a nonnegative Leslie companion matrix of dimension $d + q$ whose spectrum contains all roots of C and whose Perron eigenvalue is b.*

Proof. If $p_k \leq 0$, set $a_k = -p_k \geq 0$. Then (**??**) is

$$z^{d+q} - \sum_{k=1}^{d+q} a_k z^{d+q-k},$$

the characteristic polynomial of a nonnegative Leslie companion matrix. Since $C(b) = 0$, the product also vanishes at b. The vector

$$(b^{d+q-1}, b^{d+q-2}, \dots, b, 1)^T$$

is strictly positive and is an eigenvector with eigenvalue b. The Collatz–Wielandt characterization therefore gives spectral radius b.

Conversely, if a nonnegative Leslie characteristic polynomial contains all roots of C, polynomial division by the monic factor C gives a real monic complementary polynomial R_q, and nonnegativity of the Leslie coefficients is exactly the condition $p_k \le 0$. □

Set $c_0 = 1$ and $c_k = 0$ outside $0 \le k \le d$, and set $r_0 = 1$. Then

$$p_k = \sum_{j=0}^{q} r_j c_{k-j} = c_k + \sum_{j=1}^{q} c_{k-j} r_j. \tag{29.5}$$

Define the $(d+q) \times q$ Toeplitz matrix

$$(A_q)_{kj} = c_{k-j}, \qquad 1 \le k \le d+q, \quad 1 \le j \le q, \tag{29.6}$$

and the vector

$$h_q = (-c_1, \dots, -c_d, 0, \dots, 0)^T \in \mathbb{R}^{d+q}. \tag{29.7}$$

Theorem 29.3. *For fixed q, a degree-q auxiliary completion exists if and only if the linear system*

$$\boxed{A_q r \le h_q} \tag{29.8}$$

is feasible for $r = (r_1, \dots, r_q)^T$. Consequently,

$$\boxed{\nu(C) = \min\{q \ge 0 : A_q r \le h_q \text{ is feasible}\}.} \tag{29.9}$$

Proof. Equation (**??**) gives $p_k \le 0$ exactly as the kth inequality in (**??**). Definition **??** then gives (**??**). □

Proposition 29.4 (Dual infeasibility certificate)**.** *Degree q is impossible if and only if there exists a vector $y \in \mathbb{R}^{d+q}$ satisfying*

$$\boxed{y \ge 0, \qquad A_q^T y = 0, \qquad h_q^T y < 0.} \tag{29.10}$$

Such a y is a Farkas certificate that no choice of q auxiliary roots can repair the rigid core.

Proof. This is Farkas' alternative applied to the system $A_q r \le h_q$. □

Proposition 29.5. *If degree q is feasible, every degree $q' \ge q$ is feasible. Hence the feasible auxiliary degrees form either the empty set or the tail*

$$\{\nu(C), \nu(C)+1, \nu(C)+2, \dots\}. \tag{29.11}$$

Proof. If R_q is feasible, then zR_q is monic of degree $q+1$ and

$$C(z) z R_q(z) = z\, C(z) R_q(z).$$

Its nonleading coefficients are the previous nonpositive coefficients shifted by one place, together with a final zero. Iteration proves the claim. □

29.1. **Auxiliary defect profile.** For a quantitative profile before exact feasibility, define

$$E_q(C) = \min_{r\in\mathbb{R}^q,\ e\geq 0} \{e : A_q r - h_q \leq e\mathbf{1}\}. \tag{29.12}$$

Equivalently, E_q is the minimum possible largest positive coefficient of the completed product. To compare different b values, the same definition may be applied after the sign-preserving scaling $z = \sqrt{b}\, w$, which replaces c_k by $c_k/b^{k/2}$.

Proposition 29.6. *The auxiliary defect profile satisfies*

$$E_q(C) \geq 0, \qquad E_{q+1}(C) \leq E_q(C), \qquad E_q(C) = 0 \Longleftrightarrow q \geq \nu(C). \tag{29.13}$$

Thus $q \mapsto E_q(C)$ records how the rigid obstruction is removed as auxiliary degrees are introduced, while $\nu(C)$ is its first zero.

Proof. Nonnegativity is built into (**??**). The construction $R_q \mapsto zR_q$ used in Proposition **??** preserves the coefficient violation and proves monotonicity. Finally, $E_q = 0$ is exactly feasibility of (**??**). □

29.2. **Exact one-auxiliary test.** Because the rigid core has

$$c_d = -b^{K+1} < 0, \tag{29.14}$$

a degree-one multiplier $R_1(z) = z + t$ must satisfy $t \geq 0$. Its remaining conditions are

$$c_1 + t \leq 0, \qquad c_k + tc_{k-1} \leq 0, \quad 2 \leq k \leq d. \tag{29.15}$$

Hence $\nu(C) = 1$ exactly when the interval of t values determined by (**??**) and $t \geq 0$ is nonempty. Explicitly, every index with $c_{k-1} < 0$ supplies a lower bound

$$t \geq -\frac{c_k}{c_{k-1}}, \tag{29.16}$$

every index with $c_{k-1} > 0$ supplies an upper bound

$$t \leq -\frac{c_k}{c_{k-1}}, \tag{29.17}$$

and $c_{k-1} = 0$ requires $c_k \leq 0$. The first coefficient supplies $t \leq -c_1$.

Remark 29.7. The rigid margin Δ_K and the completion index ν measure different properties. The sign of Δ_K decides $q = 0$ feasibility, but its magnitude need not determine how many auxiliary degrees are necessary. The appropriate comparison is therefore not only $-\Delta_K$ versus ν, but the full scaled defect profile

$$E_0, E_1, \ldots, E_\nu = 0.$$

For the eight $K = 7$ configurations in Experiment **??**, this profile can be computed from the already estimated phase sets without any new frequency estimation.

Experiment 29.8. For each of the eight evaluable $K = 7$ phase sets, form the fixed core polynomial $C = P_{b,\Omega}$ once. Starting at $q = 0$, solve (**??**) in increasing degree. For every infeasible q, retain a dual vector satisfying (**??**); at the first feasible degree retain the multiplier coefficients, the completed Leslie coefficients, the minimum coefficient, the primal residual, and the spectral residual. Report

$$(b, \Delta_7, j_*, \nu, E_0, E_1, \ldots, E_\nu)$$

for every case. This protocol certifies the minimal auxiliary repair degree rather than merely producing one high-dimensional feasible completion.

30. Static completion versus data-compatible repaired recurrence

The completion problem of Section **??** starts from a prescribed critical core $C = P_{b,\Omega}$ and asks whether multiplication by a monic real polynomial can make all nonleading coefficients nonpositive. This is an exact static inverse-spectral problem. For a moving scalar sequence one needs an additional condition: the completed polynomial must also be the recurrence polynomial of the data.

Fix $K \geq 1$, an auxiliary degree $q \geq 0$, and $b > 1$, and put

$$D = 2K + 1 + q. \tag{30.1}$$

For an order-D scalar recurrence define

$$h_t^{(D)} = (u_{t-1}, \dots, u_{t-D}) \in \mathbb{R}^D, \tag{30.2}$$

and for a window beginning at n set

$$\mathsf{A}_{n,D} = \begin{pmatrix} h_{n+D}^{(D)} \\ \vdots \\ h_{n+2D-1}^{(D)} \end{pmatrix}, \qquad \mathsf{g}_{n,D} = \begin{pmatrix} u_{n+D} \\ \vdots \\ u_{n+2D-1} \end{pmatrix}. \tag{30.3}$$

Definition 30.1 (Data recurrence polynomial)**.** If $\det \mathsf{A}_{n,D} \neq 0$, define

$$a_n^{(D)} = \mathsf{A}_{n,D}^{-1} \mathsf{g}_{n,D} \tag{30.4}$$

and

$$P_{n,D}(z) = z^D - \sum_{\ell=1}^{D} a_{n,\ell}^{(D)} z^{D-\ell}. \tag{30.5}$$

For $\theta = (\Omega, r) \in (0,\pi)^K \times \mathbb{R}^q$, write

$$R_q(z; r) = z^q + r_1 z^{q-1} + \cdots + r_q \tag{30.6}$$

and define the coefficient map $\Phi_{b,K,q}$ by

$$C_{b,\Omega}(z) R_q(z; r) = z^D - \sum_{\ell=1}^{D} \Phi_{b,K,q}(\theta)_\ell z^{D-\ell}. \tag{30.7}$$

Let

$$\mathcal{M}_{b,K,q}^{\mathrm{rep}} = \operatorname{im} \Phi_{b,K,q}, \qquad \mathcal{M}_{b,K,q}^{\mathrm{rep},+} = \mathcal{M}_{b,K,q}^{\mathrm{rep}} \cap \mathbb{R}_+^D. \tag{30.8}$$

Theorem 30.2 (Exact recurrence-first membership test)**.** *Assume* $\det \mathsf{A}_{n,D} \neq 0$. *The data window has an exact fixed-(b, K, q) repaired Leslie realization if and only if*

$$a_n^{(D)} \in \mathcal{M}_{b,K,q}^{\mathrm{rep},+}. \tag{30.9}$$

Equivalently, the uniquely determined recurrence polynomial $P_{n,D}$ factors as

$$P_{n,D}(z) = (z - b) \prod_{k=1}^{K} \left(z^2 - 2\sqrt{b} \cos \omega_k \, z + b \right) R_q(z; r) \tag{30.10}$$

for some real Ω, r, and its Leslie coefficients are nonnegative.

Proof. Nonsingularity gives a unique order-D coefficient vector satisfying the D recurrence equations. A repaired fixed-(b, K, q) Leslie representation of that same window must therefore have exactly that coefficient vector. By definition of $\Phi_{b,K,q}$ this is equivalent to the factorization (**??**); Leslie nonnegativity is exactly $a_n^{(D)} \geq 0$. □

Remark 30.3 (Frequency-free root form). For exact data one may first compute $P_{n,D}$ and then test whether it contains a real root $b > 1$ and K distinct conjugate pairs with squared modulus b. The remaining quotient is the auxiliary factor. If several critical subsets are available, their choice is a branch choice and must be fixed from past information or tracked continuously; future test values may not be used to select among them.

Warning 30.4 (Static LP completion is not a data fit). The Toeplitz LP (**??**) answers whether a prescribed core C can be embedded in some larger positive Leslie spectrum. Unless the resulting product polynomial also equals $P_{n,D}$, it does not establish that the scalar data window follows that completed recurrence. This distinction is essential for every moving-window inference.

31. Geometry and codimension of the repaired family

Proposition 31.1 (Auxiliary degree does not remove the critical codimension)**.** *At a point θ where the Jacobian $D\Phi_{b,K,q}(\theta)$ has full column rank $K+q$, the fixed-b repaired family is locally a smooth immersed manifold of dimension*

$$K + q \tag{31.1}$$

inside $\mathbb{R}^D$, and hence has codimension

$$D - (K + q) = K + 1. \tag{31.2}$$

The codimension is independent of the auxiliary degree q.

Proof. The parameter domain has $K + q$ real coordinates. Full column rank gives local image dimension $K + q$ by the constant-rank theorem. Substituting $D = 2K + 1 + q$ gives $D - (K + q) = K + 1$. □

Remark 31.2. Each auxiliary coefficient adds one parameter and simultaneously raises the ambient recurrence degree by one. Therefore increasing q does not make a generic scalar window automatically compatible with a critical Leslie core. Auxiliary completion repairs a positivity obstruction; it does not erase the $K + 1$ fixed-b spectral constraints.

Warning 31.3 (Fixed degree is part of the state model). The repaired mobility theorem below assumes the same q on adjacent windows. Allowing q to change alters the recurrence dimension and can create rank-deficient or nonminimal representations. A dynamic protocol must therefore fix q before test, or specify a separate causal dimension-changing theory. Silent per-window reselection of q is an additional information channel, not part of the theorem.

32. Exact repaired overlap and nonlinear scalarization

Consider two order-D recurrence windows beginning at n and $n + s$, with $1 \le s \le D$. They share exactly $D - s$ recurrence equations. Define

$$O_{n,s}^{(D)} = \begin{pmatrix} h_{n+s+D}^{(D)} \\ \vdots \\ h_{n+2D-1}^{(D)} \end{pmatrix} \in \mathbb{R}^{(D-s)\times D}. \tag{32.1}$$

Let the old coefficient vector be $a_n = \Phi_{b,K,q}(\theta_n) \in \mathcal{M}_{b,K,q}^{\mathrm{rep},+}$. Every candidate next coefficient vector on the same repaired branch that satisfies all common equations must obey

$$F_{n,s}(\theta) := O_{n,s}^{(D)} \left(\Phi_{b,K,q}(\theta) - a_n\right) = 0. \tag{32.2}$$

Theorem 32.1 (Local repaired mobility dimension)**.** *Assume $D\Phi_{b,K,q}(\theta_n)$ has rank $K+q$ and that $DF_{n,s}(\theta_n)$ has maximal possible rank $\min\{K+q, D-s\}$. Then the local solution set of* (**??**) *through θ_n has dimension*

$$\boxed{\max\{0, s-K-1\}.} \tag{32.3}$$

Proof. The parameter space has dimension $K+q$. The constant-rank theorem gives local level-set dimension $K+q-\min\{K+q, D-s\}$. Since $D = 2K+1+q$, this equals $\max\{0, s-K-1\}$. □

Corollary 32.2 (Repaired critical stride)**.** *Under the hypotheses of Theorem* **??***, the first generic one-dimensional repaired motion occurs at*

$$\boxed{s_* = K+2,} \tag{32.4}$$

independently of q.

Remark 32.3. For $q = 0$ and fixed b, the reciprocal–Perron affine hull makes the critical motion the exact line $a + \tau v$ of Section **??**. For $q > 0$, the repaired coefficient family is generally nonlinear. The one-dimensional object supplied by Corollary **??** is therefore a local curve, not in general an affine line. Replacing that curve by its tangent is only a first-order approximation.

At the repaired critical stride define the first genuinely new index

$$t_0 = n + 2D. \tag{32.5}$$

All entries of $h_{t_0}^{(D)}$ are known before u_{t_0} is revealed.

Theorem 32.4 (One observed scalar selects the local repaired continuation)**.** *Assume the hypotheses of Corollary* **??***, and let Γ_n be the regular one-dimensional local solution branch of* (**??**) *through θ_n. Define*

$$\chi_n(\theta) = h_{t_0}^{(D)} \Phi_{b,K,q}(\theta), \qquad \theta \in \Gamma_n. \tag{32.6}$$

If the differential of χ_n does not vanish on the tangent line $T_{\theta_n}\Gamma_n$, then χ_n is locally one-to-one on Γ_n. Consequently, every sufficiently near next realization on the same branch is uniquely selected by the single observed value u_{t_0}. A canonical scalar innovation is

$$\boxed{\xi_n = u_{t_0} - h_{t_0}^{(D)} a_n.} \tag{32.7}$$

Locally there is a unique map $\theta^+ = \Theta_n(\xi_n)$ and hence

$$a^+ = \Phi_{b,K,q}(\Theta_n(\xi_n))\,. \tag{32.8}$$

Proof. The regular common-equation set is a one-dimensional manifold by Corollary **??**. A nonzero derivative of the scalar map χ_n along its tangent makes χ_n a local coordinate by the one-dimensional inverse function theorem. The observed value u_{t_0} therefore selects one nearby point on the branch. Equation (**??**) is a translated scalar coordinate. □

Corollary 32.5 (Repaired $K+1$-value causal forecast)**.** *Under Theorem* **??***, assume the true next window lies on the selected local branch and uses the coefficient vector a^+ of* (**??**)*. After revealing only u_{t_0}, the remaining*

$$s_* - 1 = K+1 \tag{32.9}$$

new values of the next window are determined recursively by the order-D recurrence with coefficient vector a^+.

Warning 32.6 (Local versus finite-step continuation). Theorem **??** is local. A finite transition must additionally certify that the continuation stays on one regular connected branch, remains nonnegative, preserves the chosen critical factor, and does not cross a point where the overlap rank or the observation derivative degenerates. At such an event the correct status is `NOT_APPLICABLE` or `NUMERICALLY_AMBIGUOUS`, not branch reselection from future data.

33. Causal state and hidden-information audit

Proposition 33.1 (Rigid direction is past-determined)**.** *In the rigid fixed-b case at $s_* = K + 2$, suppose* $\ker Z_{n,s_*}$ *is one-dimensional and $h_{t_0} v \neq 0$. The normalization*

$$\|v_n\|_2 = 1, \qquad h_{t_0} v_n > 0 \tag{33.1}$$

makes v_n a deterministic function of data available before u_{t_0}. Its variation from one window to another is therefore not, by itself, an additional external information channel.

Proof. Both O_{n,s_*} and R_b are determined before u_{t_0}, hence so is the one-dimensional nullspace. Unit norm removes scale and the sign condition removes the remaining sign ambiguity. □

Proposition 33.2 (Auxiliary coefficients are not independent innovations on a regular repaired branch)**.** *Fix K, q, b and a regular local branch as in Theorem* **??**. *Then all components of $\theta^+ = (\Omega^+, r^+)$ are locally functions of the single scalar ξ_n. In particular, the q auxiliary coefficients r^+ do not supply q independent new parameters on that transition.*

Proof. Theorem **??** gives the local map $\theta^+ = \Theta_n(\xi_n)$. Projection onto the r coordinates gives $r^+ = \pi_r \Theta_n(\xi_n)$. □

Warning 33.3 (When hidden channels re-enter). Proposition **??** fails as an information statement if any of the following is selected anew using unseen data:

(1) the auxiliary degree q;
(2) the Perron value b when no causal rule for b has been fixed;
(3) a disconnected or singular factorization branch;
(4) an independently optimized auxiliary multiplier in each window;
(5) a window-specific normalization, forcing term, observation operator, or state reset.

Every such choice must either be part of the past state with a deterministic update rule or be counted as an additional innovation.

Remark 33.4 (The minimum-norm LP selector). A minimum-norm solution of the static LP is a legitimate deterministic selector for one fixed core, but applying it independently to every new core does not prove that the resulting sequence is the continuation curve of Section **??**. Static canonicality and dynamic causality are different requirements.

34. Information accounting

The phrase "one innovation" is a statement about local degrees of freedom, not about a finite total amount of information in an infinite trajectory.

Proposition 34.1 (Conditional block determinacy)**.** *Let X_n denote the complete past-determined state of a regular fixed-(K, q, b) repaired transition, including the selected local branch. Under Theorem* **??**, *there is a deterministic map G_n such that*

$$(u_{t_0}, u_{t_0+1}, \ldots, u_{t_0+K+1}) = G_n(X_n, \xi_n). \tag{34.1}$$

Moreover ξ_n is itself determined by X_n and u_{t_0} through (**??**). *Hence, after X_n and the first new scalar are known, the remaining $K + 1$ values contain no further exact degrees of freedom within the model.*

Remark 34.2 (Entropy formulation). If a discrete or explicitly quantized stochastic model is imposed, exact block determinacy gives

$$H(U_{t_0:t_0+K+1} \mid X_n) = H(U_{t_0} \mid X_n)\,, \tag{34.2}$$

because the remaining block is a deterministic function of (X_n, U_{t_0}). For continuous variables, differential entropy and finite-resolution coding require additional care, so no bit count is asserted without a declared quantization or probability model.

Remark 34.3 (Unbounded temporal information remains possible). The sequence (ξ_n) can itself contain new information at every transition. Thus finite state dimension does not imply finite information for the full infinite trajectory. A law for (ξ_n) is needed for a zero-latency generative model, but it is *not* needed to test the weaker and already nontrivial claim that one revealed scalar makes the next $K+1$ values redundant.

35. Why the previous approximate scalarization is not a coherent relaxation

The exact rigid theorem assumes that the old coefficient vector already satisfies the common recurrence equations and lies in the fixed-b reciprocal–Perron affine hull. Projecting an inexact least-squares coefficient vector onto the affine hull and then moving in the exact overlap kernel does not repair a pre-existing common-equation residual.

Proposition 35.1 (Overlap residual is invariant along the rigid null direction)**.** *Let O be the common-equation matrix, let y be the corresponding right-hand side, and let $v \in \ker O$. Then for every coefficient vector a and every scalar τ,*

$$O(a+\tau v) - y = Oa - y. \tag{35.1}$$

Therefore a nonzero common-equation residual cannot be corrected by choosing τ.

Proof. Subtracting gives $\big(O(a+\tau v)-y\big) - (Oa-y) = \tau Ov = 0$. □

Corollary 35.2. *A causal forecast based on a projected or constrained least-squares base a_C with $Oa_C \neq y$ and an exact null direction $v \in \ker O$ is not a numerical relaxation of Theorem* **??***. Its forecast error may diagnose that chosen projection procedure, but it does not by itself falsify or verify the exact scalarization theorem.*

A coherent approximate model must instead be posed directly on the nonlinear admissible family, for example by minimizing

$$\|\mathsf{A}_{n,D}\Phi_{b,K,q}(\theta) - \mathsf{g}_{n,D}\|_W \tag{35.2}$$

using past data only, and by propagating the full set of overlap-compatible parameters satisfying a predeclared residual tolerance. If that set is not effectively one-dimensional or the prediction interval is broad, the output is ambiguous rather than a forced scalar forecast.

Remark 35.3 (Conditioning). Near a regular exact point, the smallest nonzero singular value of the overlap Jacobian controls first-order sensitivity of the recovered branch to perturbations. Consequently every approximate implementation should report the singular spectrum and stability under row/column scaling and perturbation; a binary numerical rank alone is insufficient.

36. Audit of the 5 September 2026 Mertens computation

This section records the output of `leslie_mobius_information_audit_v1_1` and reinterprets it using the corrected distinctions above. The purpose is to freeze a baseline, not to promote approximate diagnostics to theorems.

Experiment 36.1 (Frozen v1.1 baseline)**.** The observable used 161 values on a geometric grid with sampling base $B_{\mathrm{obs}} = 1.1$, split into 96 training and 65 test points. Training-only model selection over $K \in \{3,4,5,6,7\}$ and 41 Perron candidates selected

$$K = 4, \qquad m = 9, \qquad s_* = 6, \qquad b = 1.02. \tag{36.1}$$

The selected b was exactly the lower endpoint of the searched grid.

For the strict rigid test, no training or test window satisfied all required conditions. In particular the exact fixed-b reciprocal test and nonnegativity test had zero accepted windows in both partitions. Because the projected base vectors were not nonnegative, the theorem-level interval I_{adm}

containing $\tau = 0$ was not available; the reported zero interval coverage is therefore only a diagnostic of the projected procedure. The approximate projected pipeline nevertheless produced

$$\mathrm{NRMSE}_{\mathrm{Leslie,test}} = 6.3668, \tag{36.2}$$

$$\mathrm{NRMSE}_{\mathrm{persistence,test}} = 1.4657, \tag{36.3}$$

$$\mathrm{NRMSE}_{\mathrm{AR,test}} = 0.67843. \tag{36.4}$$

Leslie beat AR in none of the ten evaluated test transitions.

The numerical identity reconstructing the first new observation from the fitted scalar innovation had maximum relative discrepancy 2.689×10^{-17}, which confirms only that the scalar formula recodes the first newly observed value. The ten test innovations gave the exploratory entropy ratio

$$\widehat{H}(\tau_n \mid \tau_{n-1})/\widehat{H}(\tau_n) = 0.3142 \tag{36.5}$$

and an AR prediction ratio for τ of 1.8136 relative to the median predictor. The sample size is too small for the entropy estimate to support a law for τ.

The median angle between successive normalized rigid null directions was 64.8726°. By Proposition **??**, this large variation is not itself evidence of an extra information channel.

For the ten projected critical cores used by the approximate pipeline, no static auxiliary completion with $q \le 4$ was found, and all 40 attempted positive degrees $q = 1, \dots, 4$ had numerical Farkas certificates. This establishes a lower-degree obstruction for those *projected cores*; because those cores were not exact recurrence polynomials of the data, it does not establish $\nu > 4$ for a data-compatible repaired Mertens model.

Finally, an optional exact symbolic calculation on the unnormalized integer values $M(x)$ examined two nonsingular order-9 windows. In both, the gcd of the $K + 1$ polynomial fixed-Perron constraints was constant. For those two raw windows and $K = 4$, this excludes any b satisfying all rigid reciprocal–Perron identities.

Remark 36.2 (Null-model comparison). With the same frozen approximate model, the median test forecast errors across 100 null realizations were approximately 2.354 for support-preserving random signs and 2.464 for permuted Möbius values, both substantially below the Möbius value 6.3668. The admissible-interval fraction was zero for Möbius and both null classes. The entropy-ratio and null-direction-angle diagnostics did not produce a favorable Möbius separation. These observations are negative for that approximate protocol, not for the exact repaired theory developed in Sections **??**–**??**.

Remark 36.3 (Operational consequence). The selected $b = 1.02$ cannot be interpreted as a discovered arithmetic constant because it lies on the search boundary. Extending the numerical grid toward 1 before repairing the model would amount to further adaptation after a failed out-of-sample test. The baseline (**??**)–(**??**) should instead remain frozen for later regression testing.

37. Revised causal protocol for repaired windows

Experiment 37.1. Before any test window is examined:

1. fix K, a single auxiliary degree q, and either a fixed b or a deterministic past-only rule for b covered by a separate theorem;
2. fix the scalar observable, window convention, stride $s_* = K + 2$, numerical tolerances, branch-continuation rule, and forecast metric;
3. validate the exact algebra and continuation code on synthetic trajectories for which the true (Ω, r) path is known;
4. on each real window, construct the order-D recurrence from the data first and require the membership test of Theorem **??**;
5. reject or mark ambiguous every singular, nonpositive, branch-ambiguous, or ill-conditioned state;

(6) construct the common-equation branch from past values only;
(7) reveal only u_{t_0}, compute the canonical innovation ξ_n, and solve for the next point on the same branch without using later test values;
(8) predict the remaining $K+1$ values and compare against predeclared baselines and nulls.

A successful repaired experiment must therefore establish both *membership* and *transport*. Static completion of an externally estimated phase set is not a substitute for either step.

38. Decision criterion for the Leslie programme

Conjecture 38.1 (Data-compatible repaired trajectory)**.** *There exist fixed integers $K \geq 1$ and $q \geq 0$, together with a fixed or predeclared past-only Perron rule, such that a persistent family of Mertens windows admits data-compatible repaired Leslie recurrence polynomials in $\mathcal{M}^{\mathrm{rep},+}_{b,K,q}$ and the corresponding regular critical-stride continuations predict the remaining $K+1$ new values from one revealed scalar innovation better than predeclared null and baseline models without recalibration.*

Remark 38.2 (Falsification). For a fixed protocol, the conjecture fails if exact or tolerance-certified membership is sporadic, the auxiliary degree or Perron rule must be repeatedly retuned, branch continuation is not causal and unique, the repaired overlap fails to be effectively one-dimensional, the first-new-value map loses observability, or the out-of-sample forecast does not improve on the predeclared controls.

Remark 38.3 (What a positive result would and would not mean). Persistent success would establish a strong dynamic redundancy property of the chosen Mertens observable: conditioned on the past state, one newly revealed scalar would determine the next $K+1$ values. It would not by itself produce a zero-latency law for Mertens. That stronger goal requires an additional arithmetic law or restriction for the innovation sequence (ξ_n).

Remark 38.4 (Pre-Möbius validation gate). Before Conjecture **??** is tested again, the repaired membership, continuation, branch tracking, and forecast routines should be validated on exact synthetic examples, including cases with known auxiliary degree and deliberately introduced rank, positivity, and observability failures. Numerical outputs should distinguish `PASS`, `FAIL`, `NOT_APPLICABLE`, and `NUMERICALLY_AMBIGUOUS`.

39. Conclusions

Proposition 39.1 (Version 7 structural conclusion)**.** *The exact moving-window theory now has four distinct structural layers.*

(1) *Rigid critical Leslie spectra possess the coefficient, strain, positivity, and modal geometry developed in the earlier sections.*
(2) *For fixed b, rigid coefficient vectors obey the exact reciprocal–Perron affine constraints, which yield the affine critical-stride scalarization at $s_* = K+2$.*
(3) *Static positive completion of a prescribed rigid core is exactly the Toeplitz feasibility problem (**??**), but this completion is not automatically a recurrence of the data.*
(4) *For fixed (K, q, b), the data-compatible repaired coefficient family has regular codimension $K+1$ independently of q. Exact overlap of order-D data windows therefore has the same generic critical stride $s_* = K+2$; at that stride the repaired continuation is locally a one-dimensional curve. Under first-sample transversality, one newly observed scalar selects the local next coefficient vector and the remaining $K+1$ new values are causally determined.*

These are geometric and algebraic statements. They do not assert that Mertens windows satisfy the repaired membership conditions persistently.

Remark 39.2 (Corrected status of the numerical evidence). The rigid $K = 7$ experiment remains a valid negative statement only for its eight evaluable frequency-estimator configurations. The 5 September 2026 v1.1 Mertens audit gives a separate negative baseline for the approximate projected protocol: zero strict rigid coverage at the selected (K, b) and much worse causal forecast error than AR. Its auxiliary $q \leq 4$ obstruction applies to projected cores, not to exact data-compatible repaired recurrences. The exact symbolic gcd calculation on two raw Mertens windows is a cleaner rigid obstruction for those two windows. None of these results establishes or refutes the repaired Conjecture **??**; that conjecture requires the corrected recurrence-first protocol of Section **??**.

Remark 39.3 (Remaining arithmetic problem). If the repaired one-sample forecast succeeds persistently, Leslie geometry will have reduced each critical block to one external scalar innovation but will not have supplied a law for the innovation sequence. A zero-latency characterization therefore requires genuinely arithmetic information about (ξ_n) or an equivalent state forcing. Conversely, failure of the one-sample forecast already rules out the proposed dynamic compression without needing any theory of ξ_n.

Notebook History

Version 1.0: static rigid Leslie family and spectral comparison.
Version 2.0: positive spectral completion with auxiliary eigenvalues.
Version 3.0: moving rigid Leslie matrices on consecutive windows, corrected tangent-cone formulation, first- and second-order spectral transport, and a falsifiable forecasting protocol.
Version 4.0: exact strain coordinates, minimum Perron value, universal critical region, phase Jacobian, viability conditions, and modal amplitude transport.
Version 5.0: fixed-Perron reciprocal affine hull, exact overlap rigidity, critical stride $K+2$, scalar innovation transport, admissible innovation intervals, direct determinant tests, and the closed $K = 1$ model.
Version 6.0: negative rigid $K = 7$ margin test, exact minimal auxiliary-completion index, Toeplitz LP formulation, Farkas certificates of lower-degree impossibility, and the monotone auxiliary defect profile.
Version 7.0: recurrence-first data compatibility, separation of static completion from dynamic realization, exact local repaired mobility with codimension independent of auxiliary degree, nonlinear one-scalar continuation at $s_* = K+2$, explicit hidden-information accounting, correction of the previous approximate scalarization, and a frozen reinterpretation of the 5 September 2026 numerical audit.

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